

Subj: FW: Reply Comments [REDACTED]
[REDACTED] 7/29/00 1:35:48 PM Eastern Daylight Time

From: [REDACTED]

Reply-to: [REDACTED]

To: [REDACTED]

CC: [REDACTED]

Very much enjoyed your visit and hope you had a nice trip home.

Am attaching for your info and that of Steve, the direction that the AIAS work has now turned. As we noted in our conversations, Sachs' theory essentially completes what Einstein started. It is a unified field theory, from the spacetime approach. Electrodynamics is a part of it, so that for the very first time the interaction of gravitation and electrodynamics is in the actual theory, in a fashion where one now can speak of a model that will be usable for direct engineering. It also completely replaces the 136-year old Maxwell-Heaviside theory, dramatically extending it so to speak, because the special case of M-H theory is only approached as an asymptote, but never reached in nature.

The theory now connects it all, from beneath the quarks and gluons to the largest astronomical phenomena. This is really the theory for which Einstein was searching in his own unified field theory attempts.

It has one problem which was formidable in the old days, but is no longer such. Being entirely nonlinear, there are few closed solutions. Instead, numerical methods must usually be used as a matter of course. But for the younger scientists today, that poses no problem; computer on the desk, Mathematica etc., and they simply use the numerical methods like us old guys use a simple arithmetic pocket calculator.

A section of DOE has requested funds (\$1 M) to work out the numerical methods and do some of the major calculations. Sachs' himself some time ago used his theory to make an unparalleled calculation of the Lamb shift to great accuracy.

Anyway, $O(3)$ extended electrodynamics and Sachs' broader theory met in the middle very successfully, so that $O(3)$ turned out to be a very important subset of the overall covering Sachs unified field theory.

In my view, this work – although just beginning (the union of Sachs' theory and $O(3)$ theory) – is 50 years ahead of what is presently ongoing in university.

Just wanted to keep you posted. Because it is a unified theory, one now runs headlong into electrogravitation.

It will be very interesting to see what the AIAS theoreticians work out and publish.

Very best,
[REDACTED]

[REDACTED]

This is an interesting set of comments from [REDACTED] as usual, and to make things specific a number cruncher has to applied to the problem, to calculate precisely how electromagnetic energy changes gravitational energy. Dr. Sachs has indicated that this requires the interest of a center of learning where there are bright post graduates. If a grant comes through from DoE, [REDACTED] might become interested. [REDACTED] has kindly indicated his willingness to be a consultant and I would act as a co-PI. The irreps of the Einstein group refute Maxwell Heaviside, the latter is an asymptotic limit which is never reached. The following need to be determined:

- a) If there is a high degree of electromagnetic curvature, how does this affect gravitational curvature and mass. This would determine how to construct transport systems in which the mass of the vehicle is lessened electromagnetically. The vehicle would start going upwards if the curvature is decreased sufficiently electromagnetically.
- b) Numerous new electromagnetic effects from the Sachs equations of electromagnetism, which generalize $O(3)$ and $SU(2)$ electromagnetism.
- c) What metrics to use to start the computation, because all boils down to the metric. One can use a fitting program to reproduce a particular set of empirical data in order to determine the metric.

It ought to be borne in mind that the Sachs theory has many notable successes presumably achieved analytically, e.g. the Lamb shift without renormalization.

So it is the most precise e/m theory as well as the most general. It proves that e/m is non-Abelian under all conditions. Sachs points out in his article that if there is mass or charge anywhere in the universe, spacetime is curved. There are no point masses or point charges in the Sachs theory of e/m .

----- Forwarded Message -----

[REDACTED]

DATE: 7/28/00 6:32 PM

Thanks for your comments, [REDACTED] have also emailed their comments as well. I take stock in [REDACTED] leanings in this regard, to wit, there is always some degree of ST curvature in vacuo (keeping in mind that we all now recognize that a pristine pure vacuo devoid of everything and everything does not exist in the universe). There are close facsimilies to a pure vacuo, but never quite there. In fact, if one thinks about it a bit, one can see a logical conflict buried here. If one defines vacuo as nothing (devoid of everything), one has defined the null set, which, by definition, does not exist because it is the null set. Think about this, it's tied up in logical contradictions which can keep philosophers philosophizing all the way til death do them part. So it can be reasoned that the absence of nothing, quid pro quo, implies varying levels of ST curvature. Now, [REDACTED] you do point out that occurrences of significantly higher levels of mass-as localized anomalies in ST-are, at least, correlated with occurrences of high levels of ST curvature. As to whether there is a cause-and-effect relationship-well that discussion is reserved for another day and time. At least it can be acknowledge that correlation does exist therewith. Now, to reverse the tables, shall we look at this from the perspective in your email, [REDACTED], in which you wrote that vacuo can exist somewhere devoid of mass and can exhibit no ST curvature. You and all of us now acknowledge that energy is not possible in the absence of ST curvature. Thus, vacuo $\rightarrow$ no mass $\rightarrow$ no curvature $\rightarrow$ no energy $\rightarrow$ no change-in-mass possible (because there is no mass at all). Yet, this is the null set I describe above which now is recognized as not part of the universe. The pure vacuo null set is not part of this universe. Let us conclude that there are varying levels of vacuo never absolutely devoid of all mass. Thus, we further conclude that there are varying levels of mass $\rightarrow$ varying levels of ST curvature $\rightarrow$ varying levels of energy possible therefrom $\rightarrow$ and varying levels of change-in-mass possible.

—Original Message—

From: [REDACTED]
Sent: [REDACTED]
To: L
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Subject: RE: Review

As I read it, Sach's defines the vacuum as flat space-time in which there are no masses or charges. When there is mass or charge anywhere in the universe space-time is always curved. In princip, a change in curvature may result in change of weight as well as change in energy (allowing extraction of energy) - the two must go hand in hand. This is an important point that needs to be clarified.

I also think it important to be careful about definitions. In Sach's definition of the vacuum above it is not possible to extract energy (or cause a change in mass) because there is no curvature. Perhaps Myron could comment on this. Best, [REDACTED]

—Original Message—

From: [REDACTED] S [REDACTED]
Sent: 26 July 2000 23:00

Subject: RE: Review
Importance: High

It seems that experiments reported on energy from vacuo (S [REDACTED] etc.) show reduction in the measured weight of the hardware used in the experiments. Sometimes the hardware weighs so much that liftoff does not occur and sometimes the weight balance turns the other way with liftoff. However, I'm not sure what weight changes may have been reported relative to the patented wankle magnetic "motor". And there have been other experiments performed, for which I do not have any definitive info on, so any conclusions drawn at this point are only tentative. Finally, in conclusion, it may be valid to draw the conclusion that weight changes occur with extraction of energy from vacuo. Perhaps the two are intertwined and inextricably related and connected. [REDACTED] recent email that relates the Brazilian theory on ELF effects to the theories of GR, of mass, of force, etc., is also of interest in this discussion. Taking these considerations into account, it seems that it would be helpful for Myron to develop a manuscript that either shows or, perhaps, may not show-by mathematical derivations that start with the Sachs/Evans theories-that weight changes and energy from vacuo go hand in hand.

—Original Message—

Dr

Subject: Review

I have received a copy of a very fine review by Ranada and Trueba which I will scan in sections tomorrow, I have started work on a paper developing the Sachs theory.

In my state it is essential to keep the mind busy. All disorders have cleared up except chronic major depression, which is dangerous, but controllable. It seems that energy *ex vacuo* is very important for anti-gravity technology. This fact should emerge from a proper numerical study of the Sachs theory which unifies gravity and electromagnetism. I can only fiddle around the edges with some analytical development, but quite a lot can be done this way.

